Rhotics as recent consonants in language evolution

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This paper proposes that rhotic consonants represent a relatively recent development in the evolution of human language, supported by converging linguistic and anatomical evidence. From a linguistic perspective, rhotics display distinctive phonological behaviour, notably their frequent absence in word-initial position despite their overall high frequency. This asymmetry cannot be fully explained by articulatory or perceptual constraints alone, suggesting, in combination with other types of evidence, that the absence of word-initial rhotics reflects a former historical and evolutionary stage in language development. From an anatomical and neurobiological perspective, the production of rhotics depends on finely tuned tongue movements and control, which appear to be uniquely human and relatively recent. Taken together, these findings indicate that the emergence of rhotics is linked to recent anatomical adaptations of the adult human tongue enabling the complex motor control required for their articulation.

1. Introduction

The central claim of this paper is that non-posterior rhotic consonants (i.e., apical, alveolar, and retroflex trills, taps, and flaps, mainly denoted by means of the phonetic symbols ɾ, r, ɹ, ɽ, ɻ, ɽ of the IPA) emerged relatively recently in spoken languages, and that their development was enabled by the recent evolution of the human vocal tract, particularly the tongue. This claim is supported by typological, phonological, and historical linguistic evidence, as well as by anatomical data.

2. The linguistic evidence

2.1. The special properties of rhotics

In this study, the terms rhotic and r-sound refer to non-lateral liquid segments that are prototypically realized as taps, flaps, or trills articulated in the dental, alveolar, or pre-palatal (i.e., coronal) region, including retroflex sounds. They also denote palatalized, velarized, glottalized, lateralized, and voiceless variants of these segments, as well as the lateral /l/ in languages where this sound has a prominent rhotic allophone, for instance Korean. It excludes posterior (uvular) rhotics such as ʀ and ʁ, which are overall very rare segments in the world's languages, although their presence in two prestigious European languages, German and

French, may give a misleading impression of their frequency. It is worth noting that the two most common rhotics cross-linguistically are the apical trill [r] and the apical flap or tap [ɾ] (Maddieson 1984: 78-79).

Rhotic consonants exhibit a number of distinctive properties that set them apart from other consonants.

First, they display asymmetric distributional patterns in the world's languages because, in many languages, they tend to occur with high frequency in non-initial word positions, while being absent or rare in the word-initial position. Prior research (Labrune 2021) has established that nearly half of the languages in a balanced sample of 200 languages of the world (WALS, Dryer & Haspelmath 2013) that contain at least one phonemic liquid consonant possess few or no words beginning phonemically or phonetically with at least one of the rhotics of their inventory in their native lexicon (see also Proctor 2009; Walsh Dickey 1997, for similar observations), a feature henceforth labelled WIRA (Word-Initial Rhotic Avoidance, following Labrune 2021). Such languages include, for instance, Chechen, Basque, Japanese, Aymara, Susu, and others.

Second, the widespread restriction on word-initial rhotics in many of the world's languages cannot be solely explained by articulatory, perceptual, or functional factors. Although r-sounds are known to be difficult to articulate (Solé 2002) and are among the last segments to be acquired in language development (Kehoe 2018), there is no conclusive evidence linking these difficulties specifically to the word-initial position. Similarly, at the perceptual level, salient segments tend to be favored in word-initial positions (Blum et al. 2024; Katz & Pitzanti 2019), but it remains unclear whether rhotics are inherently salient. On one hand, they generally exhibit high sonority and longer duration (especially trills); on the other hand, they provide less acoustic contrast with the following segment (often a vowel) than most other consonants. Moreover, some apparent paradoxes emerge. Rhotics are very frequent cross-linguistically—occurring in 75 to 80% of the world's languages, and languages that possess a single liquid are more likely to have an r-sound than a lateral (Maddieson 1984:83). As already mentioned, rhotics are also highly frequent within individual languages, even in those that restrict them from the word-initial position; they are often among the most frequent consonants within a language (cf., e.g., English, French, or Japanese; see Labrune 2014 for a discussion and more examples). This is not what would be expected of a difficult sound (Zipf's law) or a late-acquired segment, as such segments are generally rare both cross-linguistically and intra-linguistically. Functional reasons, such as the preference for phonologically salient segments at word beginnings must also be considered since the word functions as an abstract organizing frame unit. In this light, rhotics could be analyzed as phonologically "weak" segments, not suited to the initial position of words where "strong" segments are preferred (Fougeron & Keating, 1997; Hyman, 1978; Lehiste, 1964; Trubetzkoy, 1939; etc.). But if this line of analysis is adopted, then, a number of other segments such as vowels and glides are also expected to be avoided word-initially. However, they are not, especially vowels.

This leads to the conclusion that articulatory and perceptual difficulty, as well as functional causes, are not crucial factors to explain WIRA.

Thirdly, it can also be demonstrated that WIRA cannot be considered the result of a frequent or natural phonological change within an evolutionary phonology approach such as that proposed by Blevins (2003). On the contrary, my data show that the overwhelming trend is for languages to evolve from a WIRA status to a non-WIRA status. In the 200-language WALS sample used for this study, no language was found to have evolved from being non-WIRA to WIRA. Outside of this sample, a few such cases have been documented, but they are extremely rare (e.g., Gascon, Walliser German). This leads me to conclude that WIRA is a recessive feature that tends to disappear over time. My data also clearly indicate that WIRA loss always results from contact with a non-WIRA language.

This view is further supported by the observation that WIRA appears particularly prevalent in proto, ancient, and extinct languages and that a number of languages whose ancestors were WIRA languages no longer display this feature. In other words, the change from WIRA to non-WIRA is common, whereas the reverse is extremely rare. WIRA is almost never acquired. It is inherited in the languages that display it. In this sense, WIRA may be characterized as a phonological fossil.

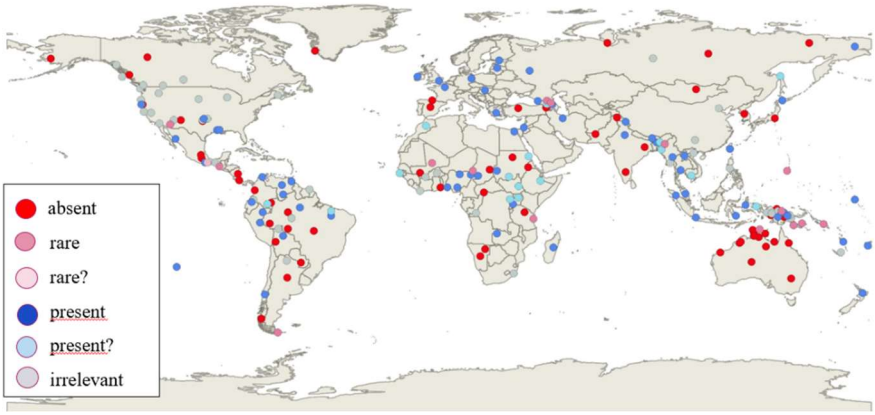

Figure 1. Geographical repartition of word-initial rhotics (200-language WALS sample)

Another notable feature of WIRA is its globally even distribution worldwide, as illustrated in Figure 1. There appears to be little evidence of geographic clustering, with the exception of mainland Australia, Europe and, perhaps, Southeast Asia.[1] This pattern suggests that WIRA may reflect the retention of a feature which was

[1] Most Australian languages are WIRA languages, and there is no evidence to suggest that this has not always been the case. In contrast, European languages of Indo-European descent are predominantly non-WIRA languages, but Proto-Indo-European has been reconstructed as a WIRA language (Benveniste, 1939). Several different language families are spoken in Southeast Asia, so the absence of WIRA in that area could be the result of chance or reflect early population and linguistic contacts leading to WIRA disappearance. But on the other hand, several of these languages are rhoticless…

present at an earlier stage, prior to the massive diffusion of the ancestors of modern languages. The geographical data in Figure 1 clearly warrant further discussion, but space constraints preclude a more detailed analysis here.

Finally, if one accepts that ontogeny may reflect, to some extent, phylogeny, then the fact that rhotics are acquired late by children, as stated above, may be seen as reflecting the fact that they emerged late in the history of human language. I will return to this issue below in section 3.

Prior to examining the anatomical evidence, let us consider the possible linguistic scenarios of rhotic emergence.

2.2. The linguistic scenarios of rhotic emergence

If one admits that rhotic consonants developed relatively late and in a secondary manner, four possible scenarios for their emergence can be considered: the lenition scenario, the hiatus-avoidance scenario, the fortition scenario, and the onset-constraint scenario. These scenarios are summarized and formalized below, where r represents any rhotic consonant, x represents any consonant other than a rhotic, V represents a vowel, 0 represents the absence of any phonetic segment.

a) Lenition scenario: x → r / V __ V (with x likely to be [d], [t], [l])
the r-sound results from the lenition of another consonant in the intervocalic environment.

b) Hiatus avoidance scenario: 0 → r / V __ V
the r-sound results from epenthesis in the intervocalic environment.

c) Fortition scenario: x → r / # __ V (with x being any consonant weaker than r)
In the word-initial position, the r-sound results from the strengthening of another consonant.

d) Onset-constraint scenario: 0 → r / # __ V
In the word-initial position, the r-sound results from the filling of an empty onset before a vowel.

Among these four scenarios, only (a) and (b) are well attested in the languages of the world, whereas (c) and (d) are not. The conclusion to be drawn is therefore as follows: rhotics initially developed in an intervocalic environment, either as the result of lenition of another consonant or through epenthesis. This alone accounts for the WIRA phenomenon.

Another subsidiary question arises: if rhotics developed late in languages, why do they occur less frequently word-initially? The answer is the following. First, it is well established that languages tend to protect word beginnings (Levelt 1989, etc.) and that they are more often suffixing than prefixing (Dryer 2020). Moreover, if rhotics originally arose through intervocalic lenition (scenario a) and epenthesis (scenario b), they would not be expected to appear word-initially at an early stage in a language's history.

The next question to be addressed is why rhotics developed relatively late. Is there a biological, cerebral, or anatomical factor that could explain this delayed development? The answer is that early *Homo sapiens* did not possess a tongue configuration like that of modern adult humans which allows for the articulation of rhotics. The anatomical evidence supporting this claim is presented below.

3. The anatomical evidence

According to Ekström & Edlund (2023), the evolution of the modern human tongue is a crucial factor in the development of spoken language. The tongue movements involved in human speech represent some of the most intricate motor actions and are believed to be unique among mammals (Sanders et al., 2013). It has even been hypothesized that the adult human tongue evolved and became specialized specifically for this purpose.

The biomechanics of tongue movements for speech differ from other tongue movements. Sanders et al. (2013) show that the adult human tongue exhibits distinctive specializations, many of which appear to be present only in humans who speak and are absent in those who vocalize without speaking, such as human neonates and macaque monkeys. More importantly, Ekström & Edlund (2023) note that current research tentatively supports Crelin's (1989) interpretation that the evolution of the human vocal tract occurred relatively recently. According to Ekström et al. (2025), our ancestors may have achieved modern vocal tract dimensions around 300 to 50 kya.

It is also noteworthy that modern humans are born with "monkey-like" vocal tracts (Sanders et al., 2013). Specifically, human infants are born with flat tongues almost entirely contained within their oral cavities, high larynges, and narrow pharynges (Crelin, 1973; and many other references), features that closely resemble those of extant non-human primates. The fine motor control of the tongue which enables adult humans to articulate apical, alveolar, and retroflex trills, taps, and flaps with precision develops only after the maturation of the tongue's muscular and nervous systems. This typically occurs by around 6 to 8 years of age, according to Denny & McGowan (2012) and other sources. Sanders et al. (2013) further note that the quantity and distribution of slow-twitch muscle fibers in the neonatal tongue closely resemble those found in their macaque specimens. These authors write that "although speculative, it is possible that this similarity reflects ontogeny recapitulating phylogeny."

One can also mention the existence of ankyloglossia, or tongue-tie, in order to illustrate the functional importance of the human tongue's fine motor control for speech. In this condition, a short or tight lingual frenulum restricts the tongue tip's mobility, sometimes impairing the production of alveolar, apical, and retroflex consonants, especially taps, flaps, and trills. This shows that even minor anatomical constraints can significantly affect articulatory precision, highlighting the tongue's specialization in human speech. Ankyloglossia thus provides insight into the delicate biomechanical adaptations that enabled the emergence of rhotics and other fine articulatory gestures (see Camargo et al., 2013; Ostapiuk, 2006, Ito et al. 2025, etc. for details).

Taken together, these biological, anatomical and developmental facts suggest that it is only recently, in evolutionary terms, that the adult human tongue reached its full modern development, enabling it to produce the extremely rapid, precise, and localized ballistic tongue-tip complex motions involved in taps, flaps, and non-posterior trills.

4. Discussion: dating the emergence of rhotics

We saw that from an anatomical and neurobiological perspective, the production of rhotics depends on finely controlled and precisely coordinated tongue movements which appear to be uniquely human and relatively recent in evolutionary terms. This observation leads to the hypothesis that rhotics are relatively recent consonants in the broader timescale of language evolution. Of course, caution is required, as the tongue is a soft tissue organ with no bones. There are no fossils that could provide a tentative dating of the various stages of tongue evolution in *Homo sapiens* or offer evidence for the subsequent emergence of rhotics. Nevertheless, and with all necessary caveats, the hypothesis that rhotics are recent consonants in the history of human speech is supported by both linguistic and anatomical evidence.

One reviewer raises the issue of a potential mismatch in time scale between the anatomical changes that enabled rhotic articulation and the rate of linguistic change, which they argue is very rapid—on the order of hundreds to thousands of years for phonetic and phonological features, if not faster. They therefore find it "very hard to believe that there would still be such a strong signature of such an old constraint in present-day or recent linguistic diversity." This objection is reasonable and merits consideration. First, with respect to rates of phonological change, the feature under discussion is a structural phonotactic property rather than an individual phoneme. It is by no means certain that word-level phonotactic patterns evolve at the same rate as individual phonemes, and it is arguably more likely that the former will exhibit greater diachronic stability.[2] Second, although the transmission rate of a given linguistic property may appear to decline rapidly over time, this is not necessarily always the case. While high levels of transmission fidelity are rare, they are nonetheless occasionally attested. Nichols (2012) estimates that a feature shared by 99% of languages at the earliest time step could exhibit a retention rate of 74% after a period of 150.000 years. Although such figures are hard to substantiate, exceptionally high levels of feature retention over deep time are theoretically possible. It is not unreasonable to assume that WIRA may have benefited from favorable conditions such as the ones discussed in section 2.2, allowing for a high rate of retention over time, especially if the period during which humans became anatomically fully capable of producing rhotic consonants is placed around 50 kya. Even if this time depth were instead set at 200 kya, the persistence of WIRA-related features in a little less than half of the world's modern languages would not be implausible. Moreover, unlike morphosyntactic features such as morphological ergativity, pronoun distinctions, or grammatical gender discussed in Nichols (2012), the initial state of rhotics total absence, and, subsequently their absence in the word-initial position (WIRA) following the scenarios discussed in section 2.2, was constrained by an anatomical

[2] While it is commonly assumed that some features are more stable than others, the causes and patterns of linguistic changes and the speed rate of these changes are not well-understood. Different languages or different aspects of a single language change in different manners and at different rates (Dediu & Cysouw, 2013).

property shared by all speakers; consequently, 100% of the initial population possessed it (except those speakers affected with ankyloglossia…).

A direct consequence of the claim advanced in this paper is that rhotics were absent from the consonant inventory of the languages spoken by the immediate ancestors of modern humans. This view thus assumes that pre-modern humans did not possess the full articulatory capacity required to produce rhotics, whereas the first modern humans did. However, the acquisition of this capacity does not necessarily entail that rhotics will be produced by those who possess it. A phonological system entirely lacking rhotics is fully viable (Maddieson, 1984: 83, reports a 23.3% rate of rhoticless languages in his sample). Rhotics may therefore have developed long after the full development of tongue dexterity in *Homo sapiens*, which renders the dating of their emergence even more difficult. Moreover, the relative chronology of consonants emergence in the human lineage remains poorly understood.[3]

Although I am unable to propose a precise date for the development of non-posterior rhotics, I tentatively suggest that they predate the Neolithic period. This proposal is based on the observation that non-posterior rhotics are cross-linguistically common overall (Maddieson, 2004:83) despite their low frequency in word-initial position. This assumption is based on the fact that the cross-linguistic frequency of a feature that does not result from regular phonetic evolution may be taken as an indicator of antiquity.

5. Conclusion

In this paper, I have argued that the most parsimonious and coherent account of the set of features, properties, and paradoxes displayed by rhotic consonants in the languages of the world is that they emerged relatively late, and that this late emergence is linked to recent anatomical adaptations of the adult human tongue, enabling the complex motor control required for non-posterior rhotic articulation.

This hypothesis has significant explanatory power: it accounts for the widespread occurrence of WIRA in modern languages despite the absence of obvious phonetic or phonological causes; it explains why WIRA behaves as a recessive feature and why it is evenly distributed across the world. It is also consistent with its presence in many ancient and reconstructed languages, as well as with its status as a linguistic fossil.

The conclusion reached in this paper is grounded in phonological reasoning and derives from a careful data-driven examination of the phonological patterning of rhotics across the world's languages, subsequently compared and crossed with evidence from anatomical evolution. Much work remains to be done. However, the hypothesis that rhotic consonants developed relatively recently in human

[3] One notable exception would be the late emergence of labiodental sounds since the Neolithic period, attributed to changes in human bite configuration, as argued by Blasi et al. (2019). Another example is /p/, a consonant described as "older than speech" by Lameira & Moran (2023). See also Everett (2021) for some proposals concerning the reconstruction of Upper Palaeolithic speech sounds.

languages as a result of anatomical changes appears to provide the only coherent explanation for the WIRA pattern as observed across the world's languages.

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